

Unified Motion Control for Multilift Unmanned Rotorcraft Systems in Forward Flight

Lucio R. Salinas^{ID}, *Member, IEEE*, Javier Gimenez, Daniel C. Gandolfo^{ID},
Claudio D. Rosales^{ID}, and Ricardo Carelli

Abstract—Interest in multilift rotorcraft systems has reemerged in recent years to overcome some of the problems in single unmanned aerial vehicle handling and delivery systems. Using multiple unmanned rotorcrafts (URs), the load weight can be evenly distributed among the vehicles, extending the flight time and therefore increasing the working distance of such systems. Moreover, with a slung-load configuration, bigger size packages with complex shape and equipment that may interfere with the onboard electronics can be carried. In this context, this article presents a payload-based unified motion control that allows any number of URs to cooperatively transport a slung load in forward flight. The motion control comprises path-following and trajectory-tracking algorithms, considers obstacle avoidance, implements a specific strategy for load weight distribution (load equalization) by regulating the relative altitude of each UR, and is robust to disturbances such as wind. In addition, this work shows that load equalization is essential for slung-load cooperative transport, especially in forward flight. The controller's stability is analytically studied and the good performance of the system is demonstrated through exhaustive simulations using validated dynamic models for mini-helicopters, cables, and payload. Finally, the relation between load weight and number of URs is analyzed by considering several scenarios.

Index Terms—Forward flight, load equalization, multilift, unified motion control, unmanned aerial vehicles (UAVs).

NOMENCLATURE

n	Number of vehicles in the formation.
ℓ	Length of the cables.
ξ_i	Position of the i th vehicle.
$\bar{\xi}$	Mean position of the formation.
ξ_ℓ	Position of the payload.

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Lucio R. Salinas is with the Department of Engineering Mathematics and the Bristol Robotics Laboratory, University of Bristol, BS8 1TH Bristol, U.K., on leave from the Instituto de Automática, Universidad Nacional de San Juan–National Scientific and Technical Research Council (CONICET), San Juan J5402CWH, Argentina (e-mail: lucio.salinas@bristol.ac.uk).

Javier Gimenez, Daniel C. Gandolfo, Claudio D. Rosales, and Ricardo Carelli are with the Instituto de Automática (INAUT), Universidad Nacional de San Juan (UNSJ)–National Scientific and Technical Research Council (CONICET), San Juan, Argentina (e-mail: jgimenez@inaut.unsj.edu.ar; dgandolfo@inaut.unsj.edu.ar; crosales@inaut.unsj.edu.ar; rcarelli@inaut.unsj.edu.ar).

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$\xi_{r,i}$	Position of the i th vehicle relative to $\bar{\xi}$.
ψ_i	Yaw angle of the i th vehicle.
$\mathbf{v}_\ell$	Payload unified motion control law.
$\mathbf{v}^*$	Payload reference velocity.
ξ_ℓ^*	Payload reference position.
ξ_ℓ^*	Trajectory-tracking payload reference velocity.
v^*	Path-following payload desired speed.
$v_{\max}^*$	Path-following payload desired maximum speed.
$k_{c,j}$	Path-following payload speed control constants.
$\mathbf{K}_{\ell,j}$	Motion control design matrices.
$\mathbf{F}_i$	Payload reaction force on the i th vehicle.
f_i	Filtered magnitude of $\mathbf{F}_i$.
f_T	Sum of all f_i .
ρ_i	Percentage of f_T assigned to the i th vehicle.
f_i^*	Desired magnitude of $\mathbf{F}_i$.
$\xi_{r,i,z}^*$	Relative altitude reference of the i th vehicle.
k_f	Load equalization relative altitude control constant.
$\xi_{r,i,xy}^*$	Desired x and y relative position of the i th vehicle.
ξ_r	Load equalization and formation task variables.
$\mathbf{v}_r$	Load equalization and formation control law.
$\mathbf{K}_{r,j}$	Load equalization and formation design matrices.
ψ^*	Desired yaw angle.
ψ	Vehicle orientation task variables.
$\mathbf{v}_\psi$	Vehicle orientation control law.
$\mathbf{K}_{\psi,j}$	Vehicle orientation design matrices.
$\mathbf{q}$	System configuration (vector of all ξ_i and ψ_i).
$\mathbf{x}$	Task variables (vector of ξ_r , ξ_ℓ and ψ).
$\mathbf{J}_\ell$	Jacobian matrix related to ξ .
$\mathbf{J}_r$	Jacobian matrix related to ξ_r .
$\mathbf{J}_\psi$	Jacobian matrix related to ψ .
$\mathbf{J}$	Jacobian matrix of the entire system.
$\mathbf{h}$	Position of the payload relative to $\bar{\xi}$.
$\dot{\mathbf{h}}$	Payload velocity in the formation frame.
$\mathbf{v}_c$	Unified motion control velocity commands.
$\tilde{\mathbf{v}}$	Velocity tracking errors.
FF_i	Light detection and ranging-based fictitious force of the i th vehicle.
$\bar{\zeta}$	Mean fictitious force in the global frame.
$\mathbf{K}_{o,j}$	Obstacle avoidance design matrices.
ξ_ℓ^{*f}	New payload reference when avoiding obstacles.

I. INTRODUCTION

THE use of unmanned aerial vehicles (UAVs), usually termed “drones,” as transportation devices has an ever-increasing number of applications [1], [2], [3]. However, the full development and deployment of these aerial transportation systems in the most promising fields, such as retail [4], medical supplies [5], and humanitarian aid [6], must still overcome several constraints: technical aspects [7] (autonomous flying and airspace integration as well as difficulties concerning battery capacity [8] and data communication); legal, moral, and ethical implications [9] (legislation, privacy, and military use); lack of acceptance by the general public [10]; and so on.

There are different aspects that can be analyzed when designing aerial load transportation systems, e.g., the type of UAV, the number of UAVs (single or multilift), and the type of payload-UAV connection (slung load or rigid links such as grasping systems). The choice of the most suitable alternative depends on the payload(s) characteristics, the mission repeatability, and the required flying distances, among others.

Regarding the type of UAV to be considered, it must be in accordance with the weight and size of the payload, along with the desired travel distance; these requirements often favor the use of fixed-wing UAVs. However, sometimes there are operational limits that lead to determine that an unmanned rotorcraft (UR) is the only viable alternative. Although these rotorcrafts have had an unquestionable advent in the last years because of their multiple advantages (superior mobility in the 3-D space, capacity to hover, and the ability to land and take off vertically in tight spaces [11], [12]), the payload characteristics often makes it impossible to use them in many applications. In order to overcome such difficulties, a cooperative transport emerges as a possible modular, scalable, and adaptable solution. Besides, a team of small URs is often less expensive and more readily available than a larger UAV with equivalent load capacity [13]. Using multiple URs, the load weight can be evenly distributed among the vehicles, extending the flight time and therefore increasing the working distance of such systems.

Unlike in single-lift UR systems, where using different kinds of grippers or manipulators to hold/manipulate a payload has some important advantages over a slung-load configuration [14], in multilift UR systems (MLURs), this type of rigid links are almost nonexistent, particularly in outdoor experimental studies. There are several reasons for the latter, these rigid systems require a significantly more precise coordination of the URs [15], their ability to change the pose and orientation of the payload is further restricted [16], and they largely modify the vehicles' dynamics (the center of mass is shifted and the pitch and roll maneuvers are technically challenging due to greater inertia) making the URs slower in terms of attitude response and less robust to perturbations [17].

These coupling effects are less noticeable if a slung-load configuration is chosen. Even though the obtained MLUR is also highly nonlinear and very hard to control, the flexible cables make it more compliant, i.e., bigger size packages with complex shape and equipment that may interfere with the onboard electronics or that may be influenced by the turbulent

flow downstream the vehicles can be more easily manipulated. Another important consideration is that the payload controllability is greatly increased and its oscillations reduced when using three or more URs, which is appealing when the payload is fragile or dangerous [17].

The concept of multilift with manned rotorcrafts such as helicopters has been around for some decades and had an important but not so successful development [18]. However, revisiting this concept now is timely because the technological advance toward fully automated aerial vehicles is providing the same hardware necessary for multilift, which is considerably reducing the extra cost of these systems [19].

In order to work with MLURs, most of the existing control systems rely on simplifying assumptions, e.g., using planar approximations [20] or ignoring the deformation and displacement of cables [21], [22], [23]. A common restriction present in every real experiment and the majority of simulated-based studies is the navigation at low speeds. Flying at near hover condition considerably disguises the mentioned challenges and allows the group of URs to use several control techniques that are not suitable for agile or forward-flight cooperative load transportation. In addition, battery capacity is a major concern for URs in general, but more so in the context of a load transportation task as every extra weight added to the onboard energy storage system decreases the vehicle's ability to carry heavier loads. This problem seriously limits the usefulness of these multilift systems and calls for higher flight speeds to reach longer distances [8], benefiting from increased flight times due to translational lift [24], [25] or a greater effective lift-to-drag ratio [26].

In MLURs, it is essential to incorporate a load equalization procedure that dynamically distributes the load weight among the vehicles; otherwise, if, e.g., all the URs navigate at the same altitude, then the vehicles in front of the formation pull the cables harder than the ones in the back making a much greater effort¹ (see Fig. 1). This effect grows critically in forward flight where the vehicles ahead are overloaded and the ones behind are doing practically zero effort.

A numeric optimization is performed in [27] to maximize the cruise performance of a multilift rotorcraft system of four identical utility helicopters. Optimal relative positions in the formation are located through simulations. Nonetheless, this information is not sufficient as it is impossible to maintain a perfect formation in high speed with intense perturbations such as wind, and small variations in the relative positions produce even bigger perturbations due to the strong coupling of the vehicles.

In [28], several alternatives for load equalization have been studied, and as in [19], [29], [30], and [31], it has been determined that regulating the relative altitudes of URs is sufficient for this purpose. Of course, this regulation process relies on sensors that can measure the cables' tension force. Sometimes, these sensors can add significant weight affecting

¹The term “effort” is used throughout this article to indicate the work required for a UR to transport the payload. In this sense, heavier payloads increase cables' tension generating greater reaction forces on the URs and consequently demand more effort from the entire fleet and from each vehicle if the load is uniformly distributed among them.

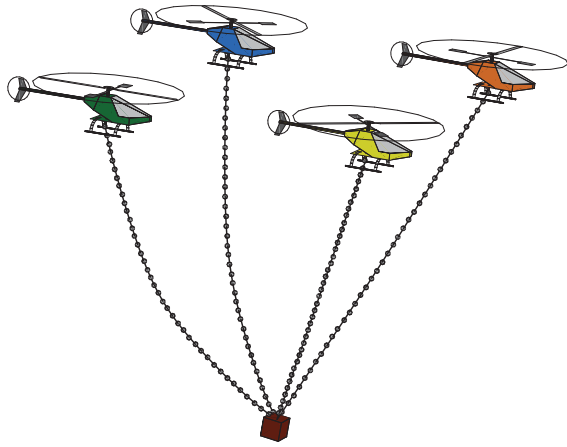


Fig. 1. Graphical representation of four mini-helicopters in a slung-load multilift configuration. All the URs are navigating at the same altitude, and then, the vehicles in front of the formation make greater effort pulling the cables much harder than the vehicles in the back. This effect is critical in forward flight regime.

the payload capacity of the URs; a possible low-cost alternative to overcome this issue is proposed in [32].

In [13], three TUB-H mini-helicopters carry a 5-kg payload (11 lb) and perform linear vertical and horizontal translation maneuvers in hover. The tension in each cable is measured and used in the closed-loop control system. Another example of an MLURS is presented in [33], where three hexacopters carry a 2.2-kg payload (4.85 lb) in hover condition except for very short periods of time. As in this study, there is no load equalization, this system is not applicable in practical tasks. In [19], two Yamaha RMAX unmanned helicopters perform several multilift outdoor trials. Although the URs do not reach a forward flight regime, they adopt speeds of around 5 m/s throughout the entire 200-s-long missions carrying a payload of up to 60 lb (27 kg), which is much faster and more challenging than any other unmanned cooperative transport experiment [17]. The authors also simulated a multilift high-speed system comprising two full-size helicopters (UH-60 Black Hawk) carrying 6000 lb (2720 kg) flying at 80 kt (41 m/s). The study shows the advantage of load equalization both in low-speed experiments and forward-flight simulations. Even though these works are very valuable because they overcome all the problems that experiments of this magnitude entail, especially [13] and [19], they only perform flights at hover or low speeds (except for one simulation), the trajectories are very simple, there are no perturbations that disrupt the formation, there are no obstacle avoidance maneuvers, and the control systems are not scalable as the number of URs is fixed.

In a preliminary work, Salinas et al. [30] modeled the problem of cooperative payload transport using URs and defined a null space-based path-following controller where obstacle avoidance was the highest priority task. The payload weight distribution according to factors such as onboard energy was also incorporated. Then, a simulation using very realistic dynamic models of a mini-helicopter, cables, and payload was made for an MLURS of four vehicles.

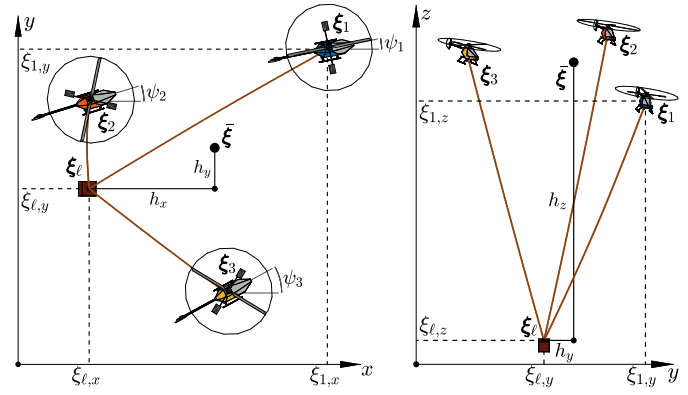


Fig. 2. System configuration and motion control task variables. Each vehicle's position ξ_i and yaw angle ψ_i along with payload position ξ_ℓ are expressed in the global frame $\langle x, y, z \rangle$. Payload position can also be defined relative to the formation mean position $\bar{\xi} = (1/n) \sum_{i=1}^n \xi_i$ as $\xi_\ell = \bar{\xi} + \mathbf{h}$, with $\mathbf{h} = [h_x, h_y, h_z]^T$.

In this article, the proposal is extended for general motion control, i.e., trajectory tracking and path following. The null space-based approach is disregarded as the obstacle avoidance task is redefined as part of the motion control, allowing a more simple unified solution. A stability proof is given without assuming perfect low-level velocity control. A formal estimation of the forward flight speed range for the URs used in the simulations is presented. The simulations are improved to achieve more realistic and practical results with higher flight velocities and cables with more resolution and greater stiffness. Finally, a comparative analysis varying the payload weight and numbers of vehicles is incorporated and the vital importance of the load equalization is highlighted with exhaustive simulations.

This article is organized as follows. Control objectives and individual control laws are specified in Section II. The inverse kinematics unified motion controller that computes all the reference velocities necessary to satisfy the stated objectives is defined and analyzed in Section III. Section IV gives the details of the dynamic modeling of the MLURS and simulation environment, defines forward-flight speeds for the simulated URs, and describes the autopilot low-level controller implementation. Simulation results are shown in Section V, and finally, a brief discussion and conclusions of the work are listed in Section VI.

II. CONTROL OBJECTIVES

Consider n URs cooperatively carrying a slung load. Each vehicle has position $\xi_i = [\xi_{i,x}, \xi_{i,y}, \xi_{i,z}]^T$, which coincides with its gravity center (g.c.) and yaw angle ψ_i (see Fig. 2). In this article, the system configuration is given by $\mathbf{q} = [\xi_1^T, \psi_1, \dots, \xi_n^T, \psi_n]^T$. In addition, consider a point-mass payload jointed to each vehicle with cables of length ℓ .

This section defines the control objectives and control laws for $\dot{\mathbf{q}}$ to generate a cooperative motion control in forward flight, while the group of URs satisfies a preset formation and properly distributes the load weight according to different factors such as lift capacity or limited onboard energy. Obstacle

avoidance is carried out by means of a reference change in the motion control and will be explained later in Section III-B.

A. Motion Control

Let $\xi_\ell = \xi_\ell(t)$ be the payload position at time t , which must follow a reference $\xi_\ell^* = (\xi_{\ell,x}^*, \xi_{\ell,y}^*, \xi_{\ell,z}^*)$. This reference can be time-indexed $\xi_\ell^* = \xi_\ell^*(t)$ if a trajectory tracking is proposed (with time requirements) or it can be parameterized by a timeless variable s [i.e., $\xi_\ell^* = \xi_\ell^*(s)$] if a path following is desired (without time requirements). These objectives together with point stabilization make up the three types of motion control studied in the literature [34]. In this work, point stabilization is not considered because the study contemplates navigation at relatively high speeds (forward flight); besides, within MLURS, this is just a special case of trajectory tracking.

Another way to represent the payload position is relative to the formation mean position $\bar{\xi}$

$$\xi_\ell = \bar{\xi} + \mathbf{h}, \quad \bar{\xi} = \frac{1}{n} \sum_{i=1}^n \xi_i \quad (1)$$

with $\mathbf{h} = [h_x, h_y, h_z]^T$ (see Fig. 2).

This article proposes a unified motion control law given by

$$\mathbf{v}_\ell = \mathbf{v}_\ell(t) = \mathbf{v}^*(t) + \mathbf{K}_{\ell,1} \tanh(\mathbf{K}_{\ell,2} \tilde{\xi}_\ell(t)) \quad (2)$$

with $\mathbf{K}_{\ell,j} > 0$, $j = 1, 2$, which solves the payload trajectory-tracking and path-following problems with a single control law, varying only the control error $\tilde{\xi}_\ell$ and reference velocity $\mathbf{v}^*$. Note that $\tanh$ is used as a componentwise vector function.

The error definition for trajectory tracking is straightforward

$$\tilde{\xi}_\ell(t) = \xi_\ell^*(t) - \xi_\ell(t)$$

while for path following is

$$\tilde{\xi}_\ell(t) = \xi_\ell^*(s(t)) - \xi_\ell(t)$$

where

$$s(t) = \arg \min_s \|\xi_\ell^*(s) - \xi_\ell(t)\|$$

i.e., $\xi_\ell^*(s(t))$ is the point on the path $\xi_\ell^*(s)$ closest to the payload position $\xi_\ell(t)$ at time t . This correspondence $s = s(t)$ varies in each experimental realization. The motion control objective is $\|\tilde{\xi}_\ell(t)\| \rightarrow 0$.

On the other hand, the reference velocity for trajectory tracking is $\mathbf{v}^*(t) = \dot{\xi}_\ell^*(t)$, while for path following, $\mathbf{v}^*(t)$ is a vector tangent to the path at $\xi_\ell^*(s(t))$ given by

$$\mathbf{v}^*(t) = v^*(t) \frac{\frac{d}{ds} \xi_\ell^*(s(t))}{\left\| \frac{d}{ds} \xi_\ell^*(s(t)) \right\|} \quad (3)$$

where $v^* > 0$ is the desired speed for the payload, which can be constant or variable. In this work, a variable speed profile similar to the one proposed in [8] is used

$$v^* = v^*(t) = \frac{v_{\max}^*}{1 + k_{c,1} \tanh(k_{c,2} \kappa(s))} \quad (4)$$

with $k_{c,i} > 0$, $v_{\max}^*$ maximum speed, and $\kappa(s)$ a function that weights the path curvature at $s(t)$ and at some point forward along the path² to reduce the desired speed during difficult-to-follow curved sections.

Finally, the unified motion control law for the payload (2) can be rewritten as

$$\mathbf{v}_\ell = \dot{\xi}_\ell^* + \mathbf{K}_{\ell,1} \tanh(\mathbf{K}_{\ell,2} \tilde{\xi}_\ell) + \boldsymbol{\varepsilon} \quad (5)$$

where $\boldsymbol{\varepsilon} = \mathbf{v}^* - \dot{\xi}_\ell^*$ for path following or null otherwise.

B. Load Equalization and Formation

1) *Load Equalization*: The equilibrium of the payload in MLURS is extremely sensitive to small changes in the vehicles' position, especially when the formation is in forward flight. The payload weight produces tensions in all the cables, and then, a noisy reaction force $\mathbf{F}_i$ with magnitude f_i acts on each UR-cable connection point.

When all the vehicles navigate at the same altitude, the vehicles in the back of the formation perform less effort because the aerodynamic forces acting on the cables and payload produce a displacement of the payload that greatly increases the tension on the cables linked to the vehicles in front of the formation and reduces the tension on the ones connected to vehicles in the back (see Fig. 1). This difference between the tensions is translated as dissimilar reaction force magnitudes f_i .

The idea to balance the cables' tension and achieve the load equalization is to find a relationship between f_i and the relative position of each UR with respect to the formation mean position

$$\xi_{r,i} = [\xi_{r,i,x}, \xi_{r,i,y}, \xi_{r,i,z}]^T = \xi_i - \bar{\xi}.$$

The easiest and quickest way for a vehicle in an MLURS to change f_i acting on it is to vary its altitude; this would of course also affect all the other reaction forces. Notwithstanding that vertical movements of the vehicles disturb the formation, other kind of movements involving horizontal displacements would only aggravate this, especially when the formation is in forward flight. When the URs are moving horizontally trying to balance the forces f_i , they can end up getting too far apart, depending on the length of the cables, and then, it could be very difficult to maintain or readjust the formation if there is a change in flying direction or a disturbance like wind gusts appears; more so as most of the URs perform pitching and rolling movements to follow horizontal velocity commands, adding delays to adjustments in the formation and distress to the UR's autopilot as the payload reaction force generates torques in the vehicle body frame that induces changes in angular accelerations ultimately affecting the same attitude angles (as shown later in Section IV-A). Moreover, vertical movements are usually very easy to perform while maintaining the same forward speed and heading for the majority of URs.

The relationship between relative altitude and f_i is dynamic and highly nonlinear. However, it can be shown that this

²Since the path is known, it is possible to use its curvature at a certain distance forward (calculated using $v_{\max}^*$ and a design time constant) to anticipate changes and at the same time further smooth the desired speed variations diminishing payload oscillations.

relationship is monotonically increasing, that is, the greater the difference between URs' altitude, the greater the difference between the URs' f_i . Hence, every vehicle can regulate its effort by only modifying its relative altitude $\xi_{r,i,z}$. Section V shows that this coordinated altitude regulation control effectively balances the load distribution, i.e., the necessary effort to transport the payload is evenly distributed among the URs. It also reveals that this is essential to avoid overloading the vehicles in front of the formation (especially in forward flight). Furthermore, although it is not presented in this study, the altitude regulation control can be tuned so that the effort that each UR makes can differ and even vary during task execution according to factors such as onboard energy and payload capacity.

Consider that $f_T = \sum_{j=1}^n \check{f}_j$ the sum of all filtered payload reaction force magnitudes (cables' tensions), and $f_i^* = \rho_i f_T$ each desired cable tension, with ρ_i as the proportion of f_T assigned to each vehicle and $\sum_{i=1}^n \rho_i = 1$. Then, the relative altitude reference for every UR is defined by

$$\xi_{r,i,z}^* = \xi_{r,i,z} + k_f \check{f}_i \quad (6)$$

with $k_f > 0$ and $\check{f}_i = f_i^* - \tilde{f}_i$. From (6), each altitude correction $\tilde{\xi}_{r,i,z} = \xi_{r,i,z}^* - \xi_{r,i,z}$ is proportional to $\check{f}_i$, where $\check{f}_i = 0$ if the cable's tension of the i th vehicle is exactly a proportion ρ_i of the total force f_T . Otherwise, if $\check{f}_i \neq 0$, the corresponding vehicle must correct its altitude in proportion to $\check{f}_i$. Then, the control objective is $\tilde{\xi}_{r,i,z}(t) \rightarrow 0 \forall i$.

2) *Formation*: Maintaining a relative position within the formation is crucial for reasons of order and security but also to minimize displacements that perturb the load equalization stage due to the MLURS strong couplings. The idea is that the URs generate a geometric pattern around the mean point ξ , while the motion control is executed. This desired pattern can be defined according to the relative position references (see Fig. 3). Since relative altitude references $\xi_{r,i,z}^*$ were defined in (6) for load equalization, only $\xi_{r,i,xy}^* = [\xi_{r,i,x}^*, \xi_{r,i,y}^*]^T$ can be set in order to define a desired formation. Thus, the formation control objective is $\|\tilde{\xi}_{r,i,xy}(t)\| \rightarrow 0 \forall i$, where $\tilde{\xi}_{r,i,xy}(t) = \xi_{r,i,xy}^*(t) - \xi_{r,i,xy}(t)$.

3) *Control Law*: To solve both load equalization and formation problems, this study proposes the following control law:

$$\mathbf{v}_r = \dot{\xi}_r^* + \mathbf{K}_{r,1} \tanh(\mathbf{K}_{r,2} \tilde{\xi}_r) \quad (7)$$

with $\mathbf{K}_{r,j} > 0$, $j = 1, 2$. In this work, the desired formation pattern is constant throughout the entire transportation task, and hence, $\dot{\xi}_r^* = \mathbf{0}$. In the case of $\dot{\xi}_{r,i,z}^*$, a simple high-gain observer without mathematical model similar to [35] and $\xi_{r,i,z}$ input is used to estimate it, followed by a low-pass filter. The components and size of ξ_r^* and $\tilde{\xi}_r$ will be further explained in Section III.

C. Vehicle Orientations

To take advantage of the benefits of forward flight in some URs such as mini-helicopters [24], the desired vehicle orientations could be selected tangent to the path or trajectory.

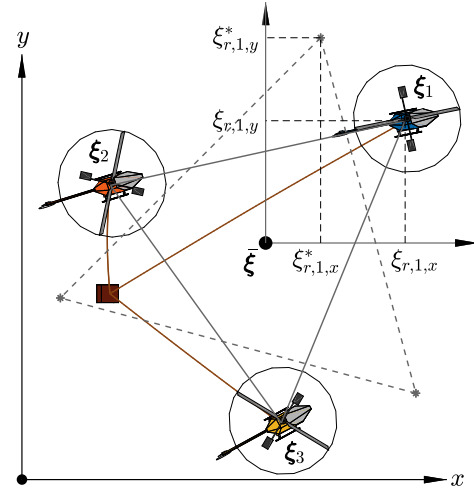


Fig. 3. Formation task variables represented with relative positions for a team of three URs. The triangular dashed lines diagram the desired geometric pattern. An axis system centered on ξ (formation frame) is depicted in which the actual relative positions $\xi_{r,i,xy}$ of the URs and the desired relative positions $\xi_{r,i,xy}^*$ are determined (only ξ_1 is indicated).

This desired yaw orientation at each point for path following is given by

$$\psi^* = \psi^*(s) = \text{atan2}\left(\frac{\partial \xi_{\ell,y}^*}{\partial s}, \frac{\partial \xi_{\ell,x}^*}{\partial s}\right)$$

and the control objective is $\tilde{\psi}_i(t) = \psi_i(t) - \psi^*(s(t)) \rightarrow 0 \forall i$. On the other hand, in trajectory tracking, the desired orientation is specified in time as

$$\psi^* = \psi^*(t) = \text{atan2}(\dot{\xi}_{\ell,y}^*(t), \dot{\xi}_{\ell,x}^*(t))$$

and the control objective is $\tilde{\psi}_i(t) = \psi_i(t) - \psi^*(t) \rightarrow 0 \forall i$.

The vehicle orientation control law has the same structure as in Section II-B3

$$\mathbf{v}_\psi = \dot{\psi}^* + \mathbf{K}_{\psi,1} \tanh(\mathbf{K}_{\psi,2} \tilde{\psi}) \quad (8)$$

with $\mathbf{K}_{\psi,j} > 0$, $j = 1, 2$. The desired yaw rate $\dot{\psi}^*$ can be easily calculated for a smooth trajectory in the trajectory-tracking problem, but in the path-following case, it is not defined as there are no time requirements. Thus, in the latter case, $\dot{\psi}^*$ is estimated using again a simple high-gain observer [35] with the desired yaw angle $\psi^*(s(t))$ as input.

III. UNIFIED MOTION CONTROL

All the objectives can be simultaneously satisfied by using an inverse kinematics unified motion controller. First, it is necessary to find the Jacobians who relate $\dot{\mathbf{q}}$ with the velocities of the task variables. These relationships can be simplified in a matrix form as

$$\dot{\mathbf{x}} = \mathbf{J}\dot{\mathbf{q}} + \mathbf{a} \quad (9)$$

where

$$\mathbf{x} = \begin{bmatrix} \xi_r \\ \xi_\ell \\ \psi \end{bmatrix}, \quad \mathbf{J} = \begin{bmatrix} \mathbf{J}_r \\ \mathbf{J}_\ell \\ \mathbf{J}_\psi \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} \mathbf{0} \\ \dot{\mathbf{h}} \\ \mathbf{0} \end{bmatrix}$$

with

$$\begin{aligned}\xi_r &= \begin{bmatrix} \xi_{r,1} \\ \vdots \\ \xi_{r,n-1} \end{bmatrix}, \quad \psi = \begin{bmatrix} \psi_1 \\ \vdots \\ \psi_n \end{bmatrix} \\ \mathbf{J}_r &= \begin{bmatrix} \Delta_{1,1} & \Delta_{1,2} & \cdots & \Delta_{1,n} \\ \Delta_{2,1} & \Delta_{2,2} & \cdots & \Delta_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \Delta_{n-1,1} & \Delta_{n-1,2} & \cdots & \Delta_{n-1,n} \end{bmatrix} \\ \mathbf{J}_\ell &= \frac{1}{n} [\Delta \quad \Delta \quad \cdots \quad \Delta] \\ \mathbf{J}_\psi &= \begin{bmatrix} \lambda_{1,1} & \lambda_{1,2} & \cdots & \lambda_{1,n} \\ \lambda_{2,1} & \lambda_{2,2} & \cdots & \lambda_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_{n,1} & \lambda_{n,2} & \cdots & \lambda_{n,n} \end{bmatrix} \\ \Delta_{i,j} &= \begin{cases} -\frac{1}{n}\Delta, & \text{if } i \neq j \\ \frac{n-1}{n}\Delta, & \text{if } i = j \end{cases}, \quad \Delta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\ \lambda_{i,j} &= \begin{cases} [0 \quad 0 \quad 0 \quad 0], & \text{if } i \neq j \\ [0 \quad 0 \quad 0 \quad 1], & \text{if } i = j. \end{cases}\end{aligned}$$

The second line of (9) is $\dot{\xi}_\ell = \mathbf{J}_\ell \dot{\mathbf{q}} + \dot{\mathbf{h}}$, which expresses the payload velocity in the global frame as the sum of two terms. The first one is the velocity of the mean point of the formation $\dot{\xi}$. The second one is the payload velocity in the formation frame, and it is defined by $\dot{\mathbf{h}} = \dot{\xi}_\ell - \dot{\xi}$. The latter contains the dynamic disturbances and all the information about the geometry of the MLURS, including the length of the cables. $\dot{\mathbf{h}}$ is very difficult to measure in practice, and thus, it is estimated using another simple high-gain observer similar to the ones used in Section II, with $\mathbf{h}$ (1) as input, followed by a low-pass filter.

Note that $\xi_{r,n}$ is not included in the vector ξ_r since $\xi_{r,n}$ and $\dot{\xi}_{r,n}$ are functions of $\xi_{r,1}, \dots, \xi_{r,n-1}$ and their velocities since $\sum_{i=1}^n \xi_{r,i} = \mathbf{0}$. Thus, $\mathbf{J}$ and $\mathbf{x}$ are arrays with sizes $4n \times 4n$ and $4n \times 1$, respectively. Finally, consider the following vectors:

$$\begin{aligned}\mathbf{x}^* &= \begin{bmatrix} \xi_r^* \\ \xi_\ell^* \\ \psi^* \end{bmatrix}, \quad \xi_r^* = \begin{bmatrix} \xi_{r,1}^* \\ \vdots \\ \xi_{r,n-1}^* \end{bmatrix}, \quad \psi^* = \begin{bmatrix} \psi_1^* \\ \vdots \\ \psi_n^* \end{bmatrix} \\ \tilde{\mathbf{x}} &= \begin{bmatrix} \tilde{\xi}_r \\ \tilde{\xi}_\ell \\ \tilde{\psi} \end{bmatrix}, \quad \tilde{\xi}_r = \begin{bmatrix} \tilde{\xi}_{r,1} \\ \vdots \\ \tilde{\xi}_{r,n-1} \end{bmatrix}, \quad \tilde{\psi} = \begin{bmatrix} \tilde{\psi}_1 \\ \vdots \\ \tilde{\psi}_n \end{bmatrix}.\end{aligned}$$

With these notations and using (5), (7), and (8), the unified motion control considering load equalization, formation, and orientation objectives can be performed by using the following inverse kinematics control law:

$$\begin{aligned}\mathbf{v}_c &= \mathbf{J}^{-1} \begin{bmatrix} \mathbf{v}_r \\ \mathbf{v}_\ell - \dot{\mathbf{h}} \\ \mathbf{v}_\psi \end{bmatrix} = \begin{bmatrix} \dot{\xi}_r^* + \mathbf{K}_{r,1} \tanh(\mathbf{K}_{r,2} \tilde{\xi}_r) \\ \dot{\xi}_\ell^* + \mathbf{K}_{\ell,1} \tanh(\mathbf{K}_{\ell,2} \tilde{\xi}_\ell) + \boldsymbol{\varepsilon} - \dot{\mathbf{h}} \\ \dot{\psi}^* + \mathbf{K}_{\psi,1} \tanh(\mathbf{K}_{\psi,2} \tilde{\psi}) \end{bmatrix} \\ &= \mathbf{J}^{-1} (\tilde{\mathbf{x}}^* + \mathbf{K}_1 \tanh(\mathbf{K}_2 \tilde{\mathbf{x}}) + \mathbf{b} - \mathbf{a})\end{aligned}\quad (10)$$

with $\mathbf{K}_j = \text{diag}(\mathbf{K}_{r,j}, \mathbf{K}_{\ell,j}, \mathbf{K}_{\psi,j})$ and $\mathbf{b} = [\mathbf{0}^\top, \boldsymbol{\varepsilon}^\top, \mathbf{0}^\top]^\top$. The existence of $\mathbf{J}^{-1}$ implies that all the tasks can be performed simultaneously without any conflict between them.

The proposed kinematic controller (10) is defined for any UR since it only proposes the desired velocities. Kinematic controllers are based on the assumption of perfect velocity tracking $\mathbf{v}_c = \dot{\mathbf{q}}$, i.e., they assume that any velocity command can be instantaneously reached. This assumption does not hold in practice due to unmodeled dynamics and disturbances. Thus, a velocity tracking error $\tilde{\mathbf{v}} = \mathbf{v}_c - \dot{\mathbf{q}}$ is introduced in the stability proof presented in Section III-A in order to contemplate these dynamics, and therefore, the stability analysis indirectly incorporates the dynamic models of the URs.

A. Stability Analysis

In this section, the closed-loop system stability is proven considering the existence of a velocity tracking error $\tilde{\mathbf{v}} = \mathbf{v}_c - \dot{\mathbf{q}}$. Then,

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{J} \dot{\mathbf{q}} + \mathbf{a} \\ &= \mathbf{J}(\mathbf{v}_c - \tilde{\mathbf{v}}) + \mathbf{a} \\ &= \dot{\mathbf{x}}^* + \mathbf{K}_1 \tanh(\mathbf{K}_2 \tilde{\mathbf{x}}) - \boldsymbol{\delta}\end{aligned}$$

with $\boldsymbol{\delta} = \mathbf{J} \tilde{\mathbf{v}} - \mathbf{b}$. Thus, the closed-loop system is given by

$$\dot{\tilde{\mathbf{x}}} + \mathbf{K}_1 \tanh(\mathbf{K}_2 \tilde{\mathbf{x}}) = \boldsymbol{\delta}. \quad (11)$$

Consider the energy function $\mathcal{V} = \tilde{\mathbf{x}}^\top \tilde{\mathbf{x}}/2$, for which

$$\dot{\mathcal{V}} = \tilde{\mathbf{x}}^\top \dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{x}}^\top (-\mathbf{K}_1 \tanh(\mathbf{K}_2 \tilde{\mathbf{x}}) + \boldsymbol{\delta}) = \tilde{\mathbf{x}}^\top (-\mathbf{g}(\tilde{\mathbf{x}}) + \boldsymbol{\delta}).$$

Both $\mathbf{g}(\tilde{\mathbf{x}}) = \mathbf{K}_1 \tanh(\mathbf{K}_2 \tilde{\mathbf{x}})$ and $\boldsymbol{\delta}$ are time-dependent vectors with $4n$ entries of the form $g_i(\tilde{\mathbf{x}}(t))$ and $\delta_i(t)$, respectively. Then, $\dot{\mathcal{V}}(t) < 0$ if $|g_i(\tilde{\mathbf{x}}(t))| > \|\delta_i\|_\infty \forall i$. If the design matrices are assumed diagonals of the form $\mathbf{K}_j = \text{diag}(k_{j,1}, \dots, k_{j,4n}) > 0$, $j = 1, 2$, then each component $\tilde{x}_i = \tilde{x}_i(t)$ of the task error vector $\tilde{\mathbf{x}}$ is ultimately bounded with bound

$$|\tilde{x}_i| < \frac{1}{k_{2,i}} \text{arctanh}\left(\frac{\|\delta_i\|_\infty}{k_{1,i}}\right)$$

which is regulated by design constants, velocity tracking performance, and $\mathbf{b}$.

B. Obstacle Avoidance

Each vehicle incorporates a 3-D light detection and ranging (LiDAR) to sense obstacles, which must be cooperatively evaded since the URs are linked to the payload.

There are two types of obstacles to be considered: internal and external. For each vehicle, the other vehicles and the payload are internal obstacles. Therefore, out-of-formation objects are considered external obstacles. The load-vehicle collisions are avoided considering that the target f_i^* on each vehicle is tracked considerably well and that the percentage ρ_i is not so small to assure that there is always tension in all the cables, whereas the collisions between vehicles are avoided by defining a desired formation shape in which the vehicles are kept sufficiently separated all the time. On the other hand, the

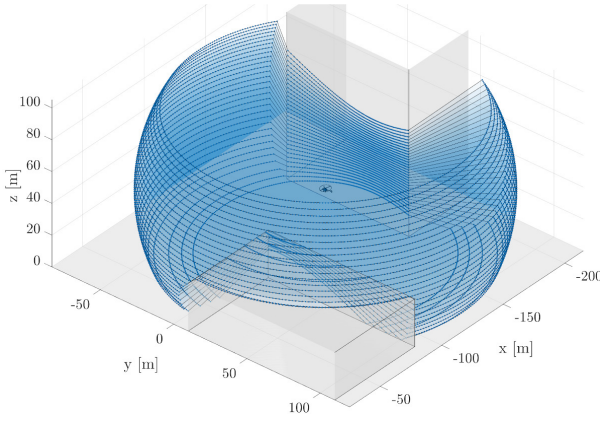


Fig. 4. Example of paths traced by a 3-D LiDAR sensor on board the mini-helicopter. The LiDAR has 360° and $\pm 30^\circ$ horizontal and vertical FOV, respectively, and 100-m range.

reference is modified in the motion control law (5) to avoid external obstacles.

Every LiDAR observation generates a fictitious force using the following expression:

$$\mathbf{FF}_i = \begin{bmatrix} F_{\text{lon},i} \\ F_{\text{lat},i} \\ F_{\text{ver},i} \end{bmatrix} = \begin{bmatrix} k_{\text{lon}} \sum_{k=1}^m D_k \cos \vartheta_k \cos \varphi_k \\ k_{\text{lat}} \sum_{k=1}^m D_k \sin \vartheta_k \cos \varphi_k \\ k_{\text{ver}} \sum_{k=1}^m D_k \sin \varphi_k \end{bmatrix} \quad (12)$$

being $F_{\text{lon},i}$, $F_{\text{lat},i}$, and $F_{\text{ver},i}$ the longitudinal, lateral, and vertical fictitious forces of vehicle i , respectively. Constants k_{lon} , k_{lat} , and k_{ver} are used to normalize the forces. $D_k = a - b d_k(t)$ indicates the magnitude of the force for each laser beam ($1 \leq k \leq m$), with $a, b > 0$ such that $a - b D_{\text{max}} = 0$ and $a - b D_{\text{min}} = 1$ for $D_{\text{min}} \leq d_k \leq D_{\text{max}}$.³ The angles ϑ_k and φ_k express the direction of each laser beam in the horizontal and vertical planes, respectively (see Fig. 4 for a graphical representation of a 360° and $\pm 30^\circ$ horizontal and vertical field of view (FOV) LiDARs, respectively).

For simplicity, each sensor is simulated as a gimbaled LiDAR (without pitch and roll inclination). Hence, $F_{\text{ver},i}$ coincides with global z -axis, and $\mathbf{FF}_i$ is rotated using ψ_i to express the fictitious force in global frame⁴ as $\zeta_i = R(\psi_i) \mathbf{FF}_i$.

The mean fictitious force $\bar{\zeta} = (1/n) \sum_{i=1}^n \zeta_i$ is then used to modify the motion reference through the following formula:

$$\dot{\xi}_\ell^{*'} = \dot{\xi}_\ell^* + \mathbf{K}_{o,1} \tanh(\mathbf{K}_{o,2} \bar{\zeta}) \quad (13)$$

where $\mathbf{K}_{o,j} > 0$, $j = 1, 2$, are diagonal design matrices.

In addition to (13), there is the capability to modify the desired x and y relative positions $\xi_{r,i,xy}^*$ to help with the avoidance. Each vehicle can use ζ_i to modify $\xi_{r,i,xy}^*$, incorporating

³The distance D_{min} can be set to avoid occlusion with the formation ignoring any reading below this value.

⁴If the LiDAR is not placed on a gimbal, then forces $F_{\text{lon},i}$, $F_{\text{lat},i}$, and $F_{\text{ver},i}$ must be filtered and the rotation matrix must include the vehicle's roll and pitch angles to properly obtain ζ_i .

flexibility when needed, by using

$$\xi_{r,i,xy}^{*'} = \xi_{r,i,xy}^*(t) + \mathbf{K}_r' \zeta_{i,xy}, \quad \mathbf{K}_r' > 0 \quad (14)$$

only if $\zeta_{i,xy}^T \xi_{r,i,xy}^* < 0$, i.e., if the relative position of each vehicle is modified in a direction that avoids an undesired reshape of the formation.

However, as this study focuses on MLURS in forward flight, to reduce the disturbances that an obstacle avoidance maneuver entails, the frontal component of $\mathbf{FF}_i$ ($F_{\text{lon},i}$) is discarded and not used in (13), and the strategy indicated in (14) is also disposed. As mentioned in Section II-B1, UR's horizontal movements greatly disrupt the formation and thus the load equalization; hence, using $F_{\text{lon},i}$ to reduce forward velocity is only desired if the obstacle sensor range is not large enough to allow an avoidance moving the formation perpendicular to the direction of motion.⁵

The motion reference change introduced with (13) also affects the payload reference velocity $v^*(t)$. In the trajectory-tracking case, the new reference velocity is obtained by differentiating (13) as follows:

$$\dot{\xi}_\ell^{*'} = \dot{\xi}_\ell^* + \mathbf{K}_{o,1} \mathbf{K}_{o,2} \text{sech}^2(\mathbf{K}_{o,2} \bar{\zeta}) \dot{\bar{\zeta}}$$

where $\mathbf{K}_{o,j}$ are diagonal matrices, sech is used as a componentwise vector function, and $\dot{\bar{\zeta}}$ is estimated using a simple high-gain observer similar to the ones used in Section II. In the path-following case, the new reference velocity is obtained following (3) calculating the vector tangent to the new path and its curvature inline.⁶

IV. SIMULATION TESTBED AND IMPLEMENTATION OF THE UNIFIED MOTION CONTROL

To develop a simulation testbed for an MLURS that mirrors a realistic scenario, several dynamic models are integrated in a custom C++ program. The URs are simulated using a nonlinear 6DoF model of a mini-helicopter [36] and the cables and payload are simulated employing point-like masses joined by springs and dampers to emulate waving and swinging movements and 3-D solid objects to calculate drag. Wind perturbations and floor collision are also incorporated in the simulation framework.

In this section, the URs and cable models are explained in more detail along with their interconnection. Then, a formal estimation of the forward flight speed range for the simulated UR is presented. Finally, a description of the unified motion control practical implementation is made.

A. Mini-Helicopter Dynamic Model

This work uses the model of a small-scale helicopter identified in [36], which is a high-fidelity nonlinear dynamic

⁵The obstacle avoidance moving the formation perpendicular to the direction of motion works well to maintain the formation in forward flight when the mission takes place in open spaces with sporadic and easy-to-detect big obstacles. In other scenarios such as cluttered environments or closed spaces, a different avoidance strategy is necessary.

⁶When the reference path changes due to an obstacle avoidance maneuver, the weight of the curvature forward along the path in function $\kappa(s)$ in (4) decreases to zero as the new path is unknown. This weight is smoothly restored as the avoidance ends and the original reference path is resumed.

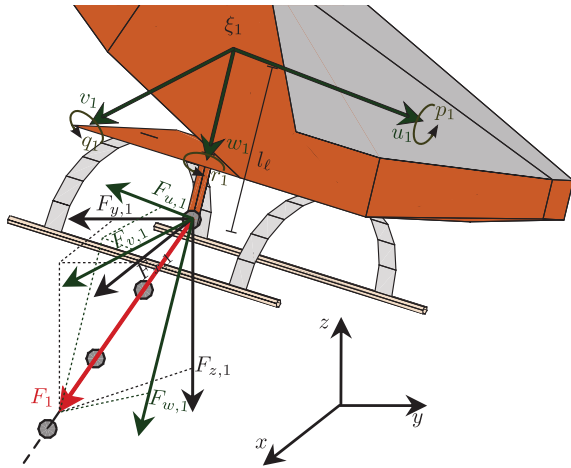


Fig. 5. Graphical representation of the mini-helicopter N° 1 and UR-cable connection point (spherical joint) below the helicopter g.c. (ξ_1). The payload reaction force (F_1) is expressed in two coordinate systems: inertial axes (x, y, z) (global frame) and helicopter's body axes (u_1, v_1, w_1) (body frame).

model that adequately represents the dynamics in hovering and up to 20-m/s forward flight, incorporating flapping, drag, and actuator dynamics.

The motion equations of a helicopter incorporating load effects are given by

$$\begin{aligned}\dot{u} &= vr - wq - gs_\theta + (X_{mr} + X_{fus})/m \\ \dot{v} &= wp - ur + gs_\phi c_\theta + (Y_{mr} + Y_{fus} + Y_{tr} + Y_{vf})/m \\ \dot{w} &= uq - vp + gc_\phi c_\theta + (Z_{mr} + Z_{fus} + Z_{ht} + \mathbf{Z}_\ell)/m \\ \dot{p} &= qr(I_{yy} - I_{zz})/I_{xx} + (L_{mr} + L_{vf} + L_{tr} + \mathbf{L}_\ell)/I_{xx} \\ \dot{q} &= pr(I_{zz} - I_{xx})/I_{yy} + (M_{mr} + M_{ht} + \mathbf{M}_\ell)/I_{yy} \\ \dot{r} &= pq(I_{xx} - I_{yy})/I_{zz} + (-Q_e + N_{vf} + N_{tr})/I_{zz}\end{aligned}\quad (15)$$

where the forces and moments acting on the helicopter are organized by components: $(\cdot)_{mr}$ for the main rotor, $(\cdot)_{tr}$ for the tail rotor, $(\cdot)_{fus}$ for the fuselage, $(\cdot)_{vf}$ for the vertical fin, $(\cdot)_{ht}$ for the horizontal stabilizer, and $(\cdot)_\ell$ for the load (cable and payload). Besides, Q_e is the torque produced by the engine to counteract the aerodynamic torque on the main rotor blades (see [36] for more details on the helicopter model and the wind effects incorporation). The payload generates a reaction force F_i on each spherical joint located $l_\ell = 0.2$ m below a helicopter g.c. (ξ_i), and this is the UR-cable connection point. Then, a force Z_ℓ and two torques L_ℓ, M_ℓ act on each UR (see Fig. 5). These magnitudes are added to the original model [see the notation highlighted with bold in (15)] and they are computed as

$$Z_{\ell,i} = F_{w,i}, \quad L_{\ell,i} = -F_{v,i}l_\ell, \quad M_{\ell,i} = F_{u,i}l_\ell \quad (16)$$

where $[F_{u,i}, F_{v,i}, F_{w,i}]^T$ is the payload reaction force components in the body frame (see Fig. 5).

The helicopter model is controlled using five input commands (C_{md}): main rotor collective and cyclic (longitudinal and lateral) blade pitch, tail rotor blade pitch, and throttle; the first four inputs control the helicopter movements (up/down, back/forth, left/right, and yaw), while the last input controls the main rotor speed. The connection between these input

TABLE I
PARAMETERS OF THE MLURS SIMULATION

Parameters	Value
Helicopter mass	8.2 [kg]
Main rotor radius	0.775 [m]
Engine max power	2000 [W]
Number of cables	1/2/3/4/5
Number of masses per cable	100
Cable link mass	0.0025 [kg]
Cable total mass	0.250 [kg]
Cable diameter	0.006 [m]
Cable drag coefficient	1.0
Payload mass	3/6/9/12/15 [kg]
Payload edge length	0.25 [m]
Payload drag coefficient	1.05
Spring length	$\approx 0.12/0.15$ [m]
Spring rate	35000 [N/m]
Spring friction	0.6 [N · s/m]
Ground repulsion*	10 [N/m]
Ground absorption*	0.1 [N · s/m]
Ground friction*	0.01 [N · s/m]
Air friction*	0.001 [N · s/m]
Air density	1.151 [kg/m ³]
Gravitational acceleration	9.7917 [m/s ²]

* Parameters for the cable (scaled up for the payload)

commands (C_{md}) and the kinematic formation control law ($\mathbf{v}_c$) is made via the low-level controller described in Section IV-D.

The simulations were performed using the nominal parameters of the MIT's X-Cell.60 acrobatic helicopter [36].

B. Flexible Cables and Payload Dynamic Models

The cables are modeled as multiple point-like masses joined by springs and dampers (Kelvin–Voigt models) to allow swinging and waving movements. This work uses 100 point-like masses for each cable (the amount of masses is set according to the desired detail level), where the first mass is attached to a spherical joint below the helicopter g.c. and the last mass is connected to a common point-like mass (payload), as shown in Fig. 1. Each mass is affected by the elastic and viscous force generated by the surrounding springs and dampers, the gravitational force, air friction force, drag force, and other ground-related forces (friction, absorption, and repulsion). The springs are all equal with a specific length, friction constant, and very high rate to avoid excessive stretching of the cable. The interaction model between the cables/payload and the ground is also parameterized. Besides, the simulation allows the variation of air density and air friction constants. The following formula is used to emulate the wind effect on the cables' links and payload:

$$F_d = \frac{1}{2} \rho c_d A v |v| \quad (17)$$

where ρ is the air density, v is the velocity with respect to air, A is the cross-sectional area, and c_d is the drag coefficient. Every link is treated as a solid cylinder and the payload is treated as a solid cube to calculate the cross-sectional area, each with its corresponding drag coefficient.⁷ Table I shows the parameters considered for the simulations.

⁷The cross-sectional area of the payload and cable is considered constant, equal to the cube face area and cylinder lateral area, respectively.

In Section V, there are several links to videos that show the dynamic behavior of the MLURS; the flexible cables' high-fidelity motion, wind perturbations, and the payload reaction force acting on each vehicle are displayed.

C. Forward Flight Condition

Rotorcraft systems fully operate from zero airspeed (unlike fixed-wing aircraft), and therefore, they present dramatic changes in their aerodynamic effects as the speed increases. The Aeronautical Design Standard (ADS-33 [37]) divides the full-size rotorcraft speeds into the following three ranges.

- 1) *Hover*: 0–15 knots (27.78 km/h or 7.72 m/s).
- 2) *Low Speed*: 15–45 knots (83.34 km/h or 23.15 m/s).
- 3) *Forward Flight*: Greater than 45 knots.

In the excellent work of Mettler [38], the effect of scale on rotorcraft dynamics is analyzed in detail. Applying Froude scaling rules, a full-size Bell UH-1H helicopter is compared to a Yamaha R-50 helicopter (a “rather big” miniature rotorcraft with 5-ft rotor radius). Following Froude scaling laws ($1/\sqrt{N}$) and using a scale ratio of $N = 4.76$, the equivalent speed ranges for R-50 are obtained.

- 1) *Hover*: 0–6.88 knots (12.74 km/h or 3.54 m/s).
- 2) *Low Speed*: 6.88–20.63 knots (38.21 km/h or 10.61 m/s).
- 3) *Forward Flight*: Greater than 20.63 knots.

Then, Mettler [38] compared R-50 with the MIT's X-Cell.60 acrobatic helicopter using Mach scaling rules.⁸ This last vehicle was identified in [36] and is the same UR model simulated in this work. When Mach scaling rules are applied, the rotor tip speed stays unchanged, and considering that both helicopters have very similar rotor tip speeds, the speed envelope for both helicopters should be roughly the same.

Consequently, to simulate the URs in forward flight, they must fly at airspeeds greater than 10.61 m/s.

D. Implementation of the Kinematic Unified Motion Control

To let the helicopters track the corresponding flight commands $\mathbf{v}_{c,i} = [\dot{x}_{i,c}, \dot{y}_{i,c}, \dot{z}_{i,c}, \dot{\psi}_{i,c}]^T$, the low-level controller stage shown in Fig. 6 is added to each vehicle.

This stage is related to the autopilot and the inner controllers used in the UR. Usually, it is composed of two steps: a rotation of velocity frame and a PID-based cascade control. In the first step, the flight commands $\mathbf{v}_{c,i}$, which are related to the inertial frame, are rotated using ψ_i to align them to the frame attached to the i th helicopter, assuming no roll or pitch movements. Then, the PID architecture generates the servo commands $C_{md,i}$ (see details in [39] and [40]).

The same unified motion control can be used for other types of UR like quadrotors by only modifying this low-level controller, following a similar procedure as in [8] and [41].

V. SIMULATION RESULTS

In this section, several realistic and challenging simulations are performed for each control type (path following and trajectory tracking), where the mini-helicopter, cables, and payload

⁸To understand why different scaling rules are applied in each comparison, please see [38].

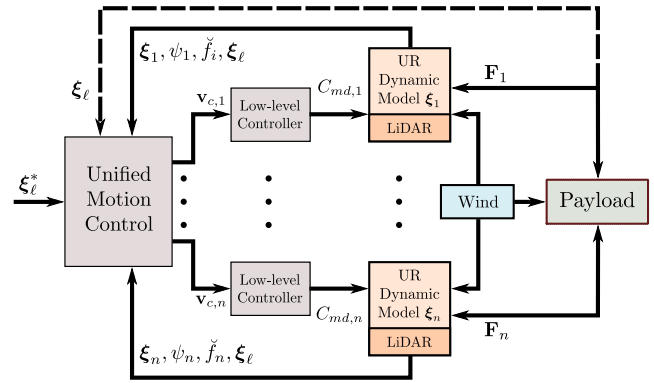


Fig. 6. Simulation framework with unified motion control for MLURS. Any number of URs can be simulated; each one connected to a common payload using a flexible cable and including its low-level controller, 3-D LiDAR, and force sensor. The wind affects every vehicle and cable, and the payload. The position of the payload can be estimated with a fusion of sensor information from each UR or with a global position sensor in the payload (dashed line).

models described in Section IV-A and IV-B are used. The commanded speed profiles fall in the forward-flight regime and wind perturbations affect all the objects (vehicles, cables, and payload).

A. Path Following

It is well known that following an upward eight-shaped curve is a complicated task; therefore, the following reference path is chosen to efficiently evaluate the controller performance:

$$\xi_\ell^* = \xi_\ell^*(s) = \begin{bmatrix} r \cos(s) \\ r \sin(2s) \\ r \frac{s}{2\pi} + 8 \end{bmatrix}, \quad \text{where } 0 \leq s \leq 2\pi \text{ and } r = 250.$$

A 3-kg payload that hangs from 12 m and 0.25 kg cables is used, and the goal is to uniformly distribute its weight between the vehicles ($\rho_i = (1/n)$). The desired formation shapes are regular polygons inscribed in a 5-m radius circle (or a pair of diametrically opposite points in the case of $n = 2$). The helicopters begin each simulation conforming the desired formation on the ground plus small random errors in position ($[-1, 1]$ m) and orientation ($[0, 2\pi]$). After the payload's takeoff, the desired navigation speed is given by (3) and (4), choosing $v_{\max}^* = 11.11$ m/s (40 km/h or 24.85 mi/h). Two obstacles are placed in the simulation environment and a constant wind of 8.5 m/s is incorporated in the path sector corresponding to $s \geq (3/2)\pi$. The wind speed corresponds to a “fresh breeze” in the Beaufort wind scale, and its direction points to the negative x - and y -axis.

Different views of the MLURS with three mini-helicopters carrying the payload are shown in Fig. 7 (a video of this simulation can be found at https://youtu.be/d3LFN_MofpI), where the formation is plotted approximately every 7 s. This figure zooms to highlight some situations. As the formation orientation does not change during the entire simulation, the load equalization system must modify the relative altitude references at every path direction change (compare zooms at t_1 and t_2). Navigation at high speeds causes the load to hang behind the formation center, but, e.g., at t_3 , a leftward

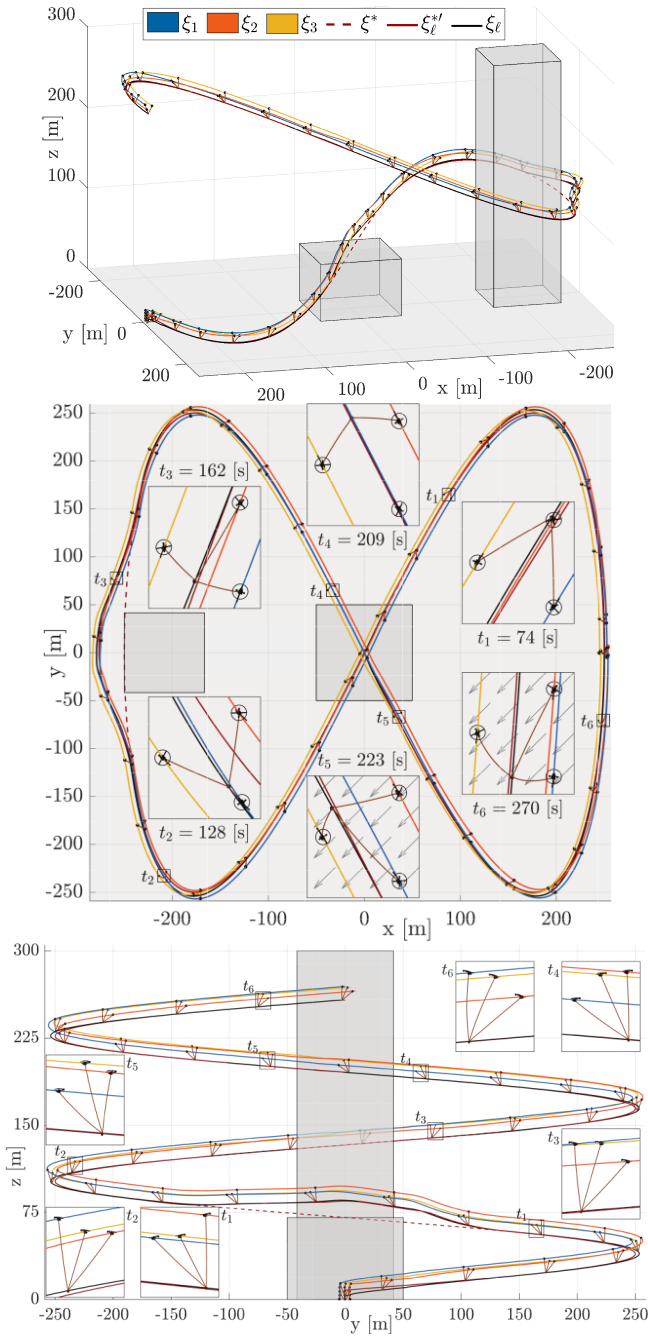


Fig. 7. 3-D view (top), XY view (middle), and YZ view (bottom) of the cooperative payload transport simulation for the path-following case. The formation is drawn approximately every 7 s.

displacement of the formation center is also observed due to the inertia caused by the obstacle side-shift avoidance. At t_4 , error-free navigation is observed prior to the sudden introduction of the aforementioned wind, which also produces a lateral disturbance (see zoom at t_5). The situations detailed at t_3 and t_5 require regulation of relative altitudes among the URs that navigate at the sides of the formation. Finally, at t_6 , the formation overcomes the combination of adversities produced by the upwind navigation at high speeds.

The path-following, formation, and orientation errors are shown in Fig. 8, along with the load equalization by

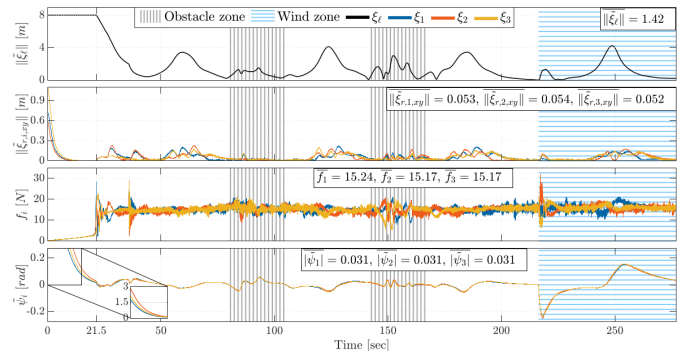


Fig. 8. Payload path-following error (top), formation errors (second), load equalization indicated by force acting on each vehicle (third), and orientation errors (bottom) obtained in the cooperative payload transport simulation for the path-following case with three URs.

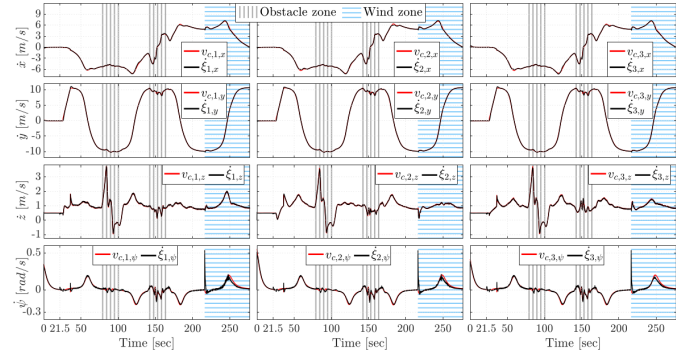


Fig. 9. Flight commands ($\mathbf{v}_{c,i} = [v_{c,i,x}, v_{c,i,y}, v_{c,i,z}, v_{c,i,\psi}]^T$) and actual vehicles' velocities ($\mathbf{q} = [\xi_{i,x}, \xi_{i,y}, \xi_{i,z}, \psi_i]^T$) for the path-following case with three URs.

the payload reaction force acting on each UR-cable connection point. The flight commands and vehicles' velocities are presented in Fig. 9. Both figures highlight the two obstacle zones and one wind zone. The first obstacle is easy to surpass increasing the formation altitude, which occurs as a consequence of a path reference change generated by the LiDAR readings (as explained in Section III-B). The second obstacle avoidance is more complex because the formation must perform side-shifts causing an increase of the formation errors and the forces produced by the payload motion. In general, path-following and formation errors are small except in curves where dynamic effects and the realistic model of the cables introduce oscillations or when the wind suddenly appears. The wind perturbs the payload much more than the vehicles due to the weight difference and the low-level control of the mini-helicopter model that compensates for it. Orientation errors are low except when the wind is suddenly introduced and in the immediate posterior turn. However, all the errors decrease rapidly. The force graphs show the need to filter these signals before incorporating them into the controller. An even distribution of the load is evidenced, except in the second obstacle avoidance and in some moments of the wind stage (e.g., the final turn). Finally, Fig. 9 shows that the low-level controller allows an almost-perfect tracking of the reference velocities generated by the unified motion control.

Fig. 10 shows a comparative analysis considering $n = 1, 2, \dots, 5$ URs. Legends show temporal means without

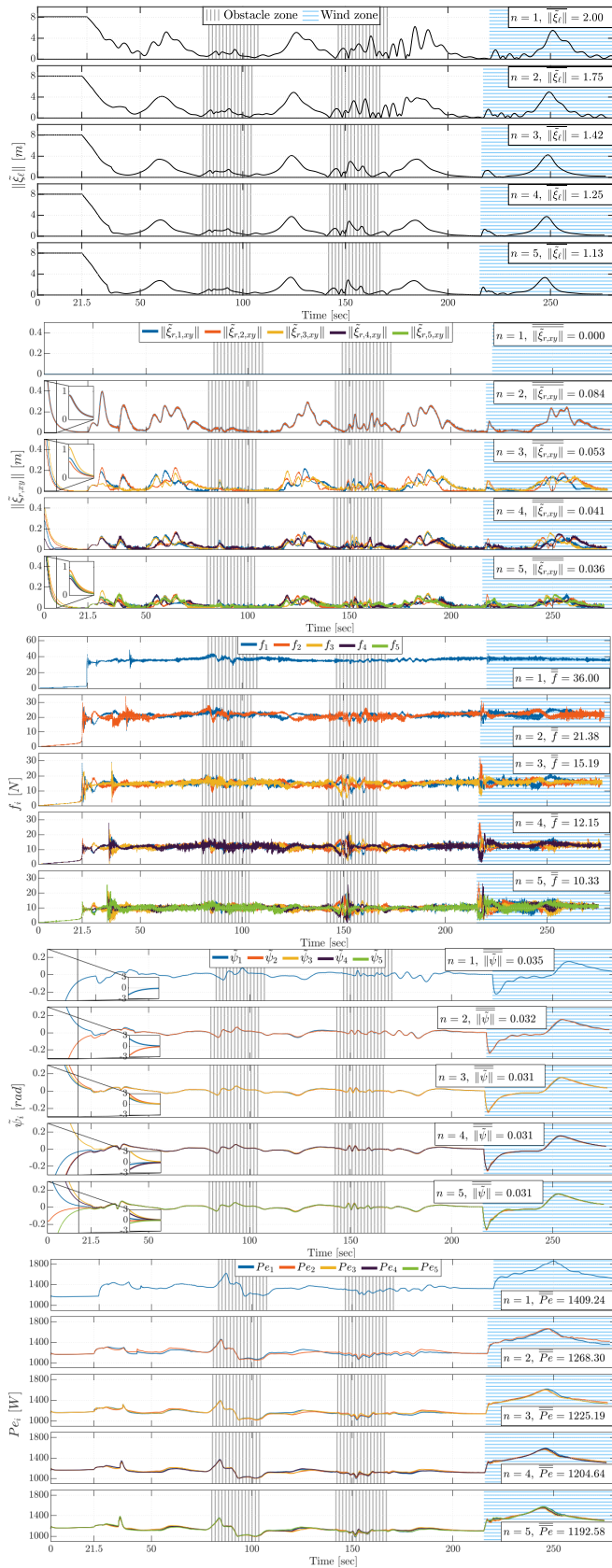


Fig. 10. Payload path-following error (top), formation error (second), load equalization indicated by force acting on each vehicle (third), orientation error (fourth), and engine power (bottom) obtained in the cooperative payload transport simulations for the path-following case with 1–5 URs and constant 3-kg (6.6 lb) payload.

considering the initial time interval in which the payload is on the ground (e.g., $t \leq 21.5$ s for $n \geq 2$) to better highlight the differences and to reduce the effects of the initial random errors. The analysis performed for three URs can be generalized to n URs. In the first plot, the path-following errors $\|\tilde{\xi}_\ell\|$ are compared, which shows the oscillation level of the payload during transport. The temporal mean of this error $\|\tilde{\xi}_\ell\|$ decreases as the number of URs increases. However, this decline is less and less noticeable, which shows the need to find a balance between the margin of error required and the extra cost of incorporating more vehicles in the MLURS. The second plot shows a similar situation with regard to the formation errors. Note that a formation error cannot be defined for $n = 1$ and that both vehicles present the same formation errors when $n = 2$. If the formation errors $\|\tilde{\xi}_{r,i,xy}\|$ of the n vehicles are averaged at each time instant, then a temporal mean of the resulting signal is calculated, and a global index of the formation errors $\|\tilde{\xi}_{r,xy}\|$ is obtained. This measure also decreases as n increases and exhibits an asymptotic behavior like $\|\tilde{\xi}_\ell\|$. The third plot illustrates the payload reaction force on the helicopters or cable tension f_i in each case and gives the global mean force $\bar{f}$ carried out during the simulation, and this index is again a temporal mean of the average force at each time instant. Although the same asymptotic decrease of previous mean errors is observed, greater force fluctuations are evident as the number of helicopters increases when obstacles or the wind introduce perturbations to the formation. The fourth plot indicates the orientation errors and its global mean $\|\tilde{\psi}\|$. These errors are practically the same for all the URs and simulations regardless of the number of vehicles; this outcome is expected as the UR-cable connection points are simulated as spherical joints. The last plot shows the engine power required by each helicopter to perform the task. The temporal mean of the average power at each time step is also calculated $\bar{P}_e$. In this case, the global mean decrease as the number of URs increases is less significant because the load weight is small compared to the helicopter's weight (see Table I). Thus, the load weight has a key role in the choice of the number of URs required to carry out the transport task without wasting energy resources. A video of these simulations can be found at <https://youtu.be/jXhMyuFnka0>.

B. Trajectory Tracking

In this section, the complexity of the desired reference for the payload is reduced in order to analyze other factors of cooperative transport. Hence, a simple trajectory as those presented in most applications is defined, i.e., a vertical takeoff stage, a straight line interrupted by sporadic obstacles, and a vertical landing stage. The trajectory smoothly accelerates after takeoff and reaches speeds of more than 12.5 m/s (45 km/h or 28 mi/h), and then, it smoothly decelerates before starting the landing procedure. In addition, a constant lateral wind of 5.55 m/s is incorporated throughout the simulation except in the take-off and landing stages. This wind starts and ends suddenly, and falls under the category of “moderate

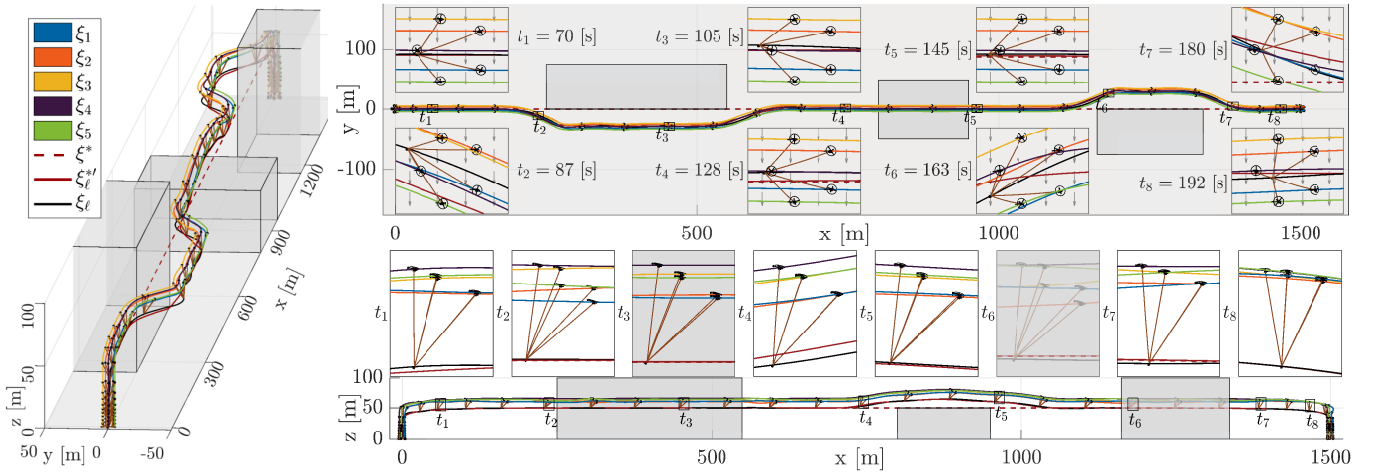


Fig. 11. 3-D view (left), XY view (right, top), and XZ view (right, bottom) of the cooperative payload transport in the trajectory-tracking case. The formation is drawn approximately every 6 s.

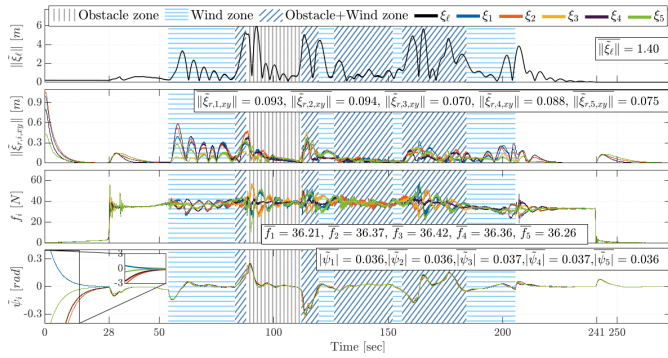


Fig. 12. Payload trajectory-tracking error (top), formation errors (second), load equalization indicated by force acting on each vehicle (third), and orientation errors (bottom) obtained in the cooperative payload transport simulations for the trajectory-tracking case with five URs.

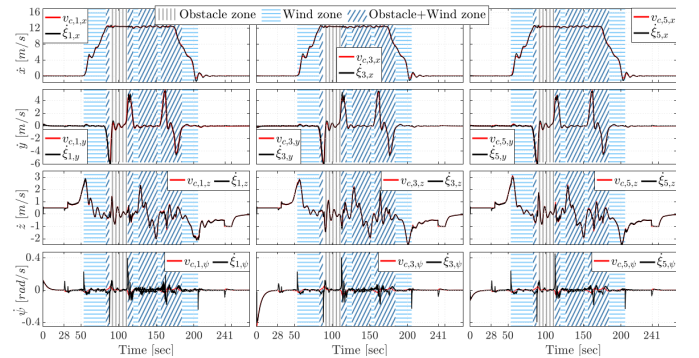


Fig. 13. Flight commands ($\mathbf{v}_{c,i} = [\dot{x}_{i,c}, \dot{y}_{i,c}, \dot{z}_{i,c}, \dot{\psi}_{i,c}]^T$) and actual vehicles' velocities ($\dot{\mathbf{q}} = [\dot{\xi}_{i,x}, \dot{\xi}_{i,y}, \dot{\xi}_{i,z}, \dot{\psi}_i]^T$) for the trajectory-tracking case with five URs (only vehicles 1, 3, and 5 are plotted).

breeze" in the Beaufort scale. In these simulations, the payload weight is proportional to the number of helicopters, that is, for each extra helicopter, the payload weight is increased by 3 kg (excluding the 0.25 kg per cable).

Different views of a formation of five helicopters carrying a 15-kg payload are shown in Fig. 11 (a video of this simulation can be found at https://youtu.be/EjOH4WL_4r8), where the formation is plotted approximately every 6 s. Besides, zooms of the formation state are included in eight key moments. At t_1 , the formation is stabilized after the sudden lateral wind; at t_2 , the avoidance of the first obstacle begins; at t_3 , the first obstacle totally blocks the wind, so the formation is not affected by it; at t_4 , the formation is shown ascending to evade the second obstacle; at t_5 , the formation is shown descending to return to the desired trajectory; at t_6 , the formation must move to the left to evade the last obstacle; at t_7 , the formation moves to the right to return to the desired trajectory; and at t_8 , the formation begins the landing.

Fig. 12 shows the trajectory tracking, formation, and orientation errors, along with the force that the load exerts on the helicopters. Again, legends show temporal means (considering only when the payload is on the air, i.e., $28 < t < 241$ s) broken down by a number of vehicles. The load stabilization

after each disturbance (especially the ones caused by the sudden wind variations) presents higher oscillation than in the path-following case. This occurs mainly by the strong influence of the wind direction (sideways) and the payload with much greater weight (five times). However, the errors remain bounded, tend to converge, and do not destabilize the system. Besides, the mean reaction forces exerted by the payload on vehicles are very similar, proving the good performance of the load equalization system.

Fig. 13 shows the similitude between flight commands (provided by unified motion control) and the actual velocities of vehicles 1, 3, and 5. The other URs present signals similar to their peers according to their relative position in the formation. Since the desired flight direction is aligned with the x -axis, the navigation velocity profile can be observed in the first row of plots, which corresponds to the forward flight regime.

A comparative analysis of errors, forces, and powers varying the number of helicopters from one to five is presented in Fig. 14. The payload weight increase in proportion to the number of vehicles produces that the decrease of the mean of the formation and trajectory-tracking errors is less than in the path-following case. Even an occasional boost in mean errors is observed for $n = 4$. The mean payload force

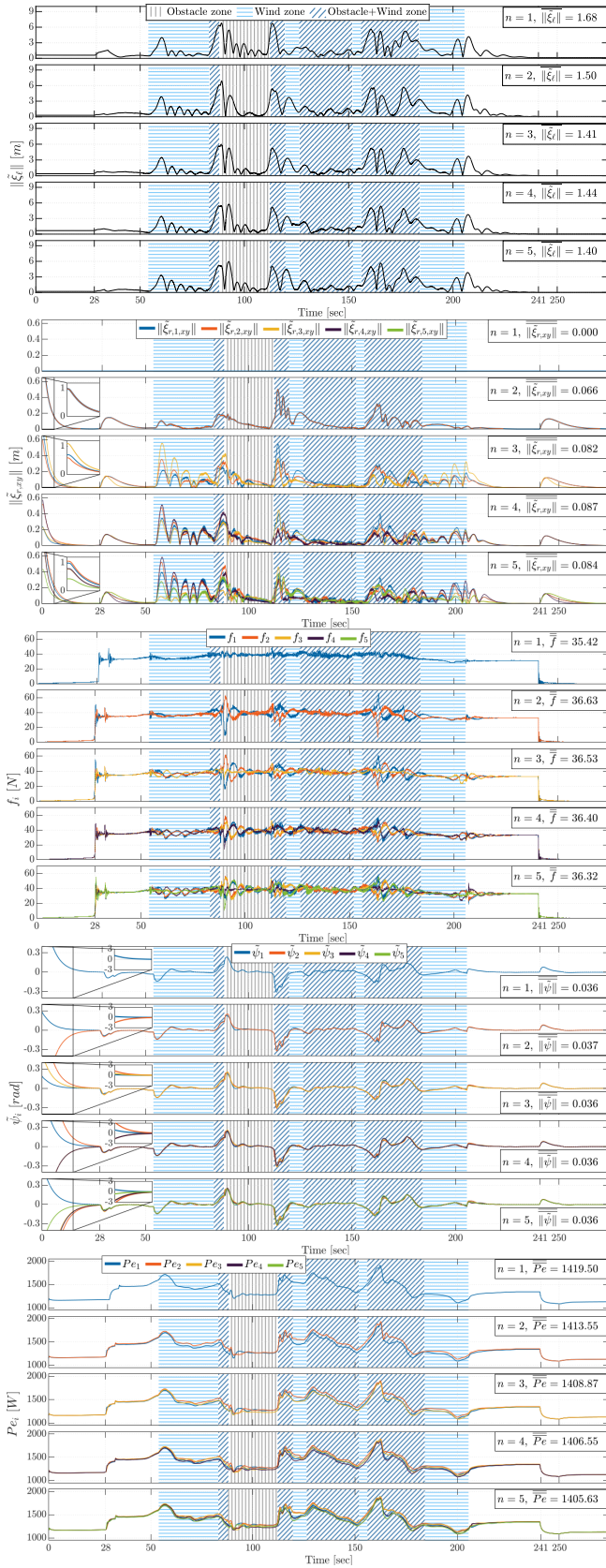


Fig. 14. Payload trajectory-tracking error (top), formation error (second), load equalization indicated by force acting on each vehicle (third), orientation error (fourth), and engine power (bottom) obtained in the cooperative payload transport simulations for the trajectory-tracking case with n (1–5) URs and $n \times 3$ kg ($n \times 6.6$ lb) payload.

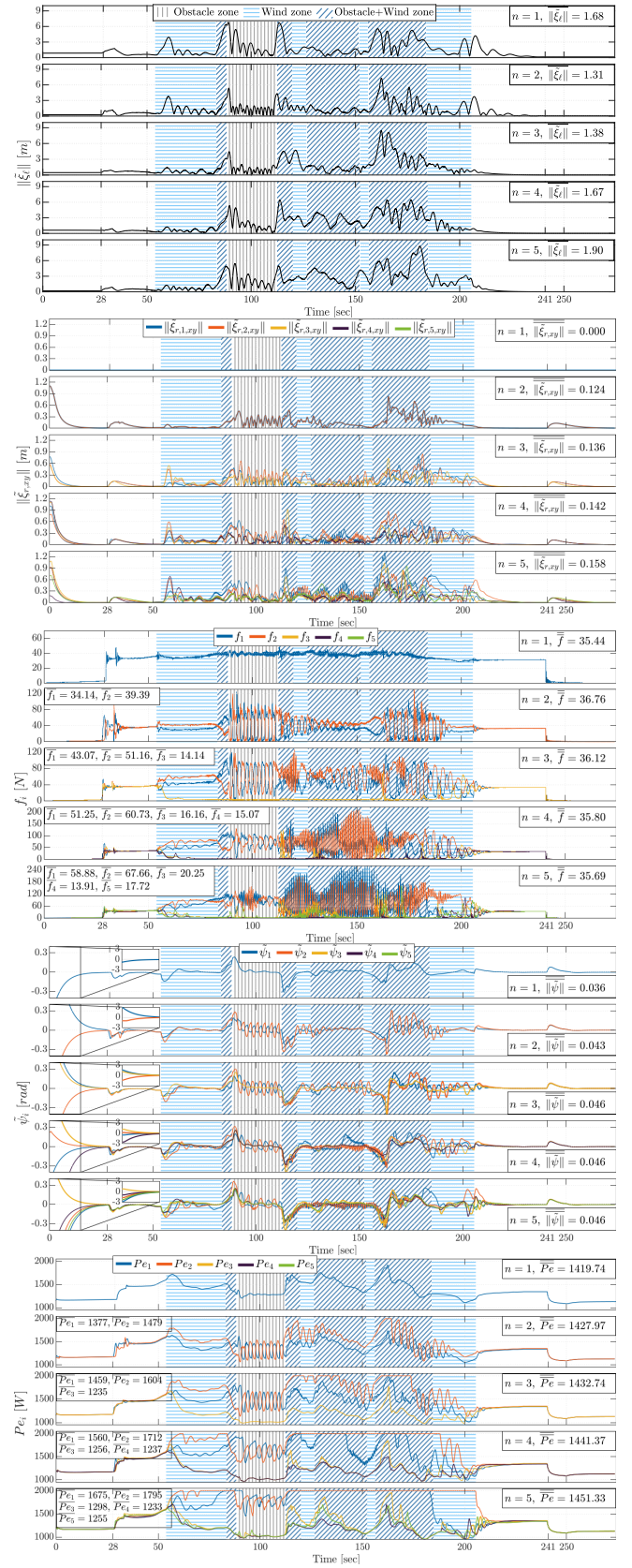


Fig. 15. Payload trajectory-tracking error (top), formation error (second), load equalization indicated by force acting on each vehicle (third), orientation error (fourth), and engine power (bottom) obtained in the cooperative payload transport simulations for the trajectory-tracking case with n (1–5) URs and $n \times 3$ kg ($n \times 6.6$ lb) payload without load equalization.

acting on the vehicles and the mean errors of orientation remain practically constant if the payload weight increases in proportion to the number of URs. The powers indicated in the last plot are smooth curves that are always below the mini-helicopter's engine maximum power (see Table I). However, the power level that each helicopter demands and the payload-helicopter weight ratio show the need to analyze whether a vehicle should be incorporated in proportion to the increase in payload weight. A video of these simulations can be found at <https://youtu.be/4lhhdDK-ods>.

C. Trajectory Tracking Without Load Equalization

In order to show how essential the load equalization system is for MLURS, the simulations carried out for the trajectory-tracking case are repeated but without a proper payload weight distribution. Instead, the formation will be controlled so that all URs navigate at the same altitude, i.e., null relative altitude references ($\xi_{r,i,z}^* = 0$) in the unified motion control. Fig. 15 shows the simulations varying the number of URs. Position, formation, and orientation errors present higher oscillations, and there is a tendency to grow as the number of vehicles increases. The most critical events can be observed in the force and power plots, where the level of oscillations shows a very unstable behavior, which can cause serious damage to the helicopters even in this simple trajectory and with moderate winds. A further proof of this is the power saturation observed for $n \geq 2$, which is due to the fact that the vehicles behind make little effort and the vehicles in front must at times bear loads that far exceed their maximum payload. Thus, MLURS without load equalization reduces the maneuverability level of the URs navigating ahead, increases energy consumption reducing flight times and therefore maximum travel distance, and can damage the vehicles or the payload or both. A video of these simulations can be found at <https://youtu.be/HYSOfCEZaM>.

VI. CONCLUSION AND DISCUSSION

This article addresses the cooperative transport of a slung load with an arbitrary number of URs in forward flight regime. A payload-based unified motion control for trajectory tracking and path following is proposed, considering obstacle avoidance, disturbances, and load equalization. The formation controller that encompasses all these considerations is based on inverse kinematics and relies on a low-level control that can handle the dynamics of the URs (and their interaction with the environment) and perform an efficient velocity control. In this article, a simple PID-based autopilot is used as the low-level control. A stability analysis is outlined and parameter-dependent error bounds are obtained. A formal estimation of the forward flight regime for the simulated mini-helicopters is presented. The MLURS is tested through exhaustive simulations based on validated dynamic models for the URs, cables, and payload.

The versatility of the proposal allows to perform simulations varying the number of URs and the type of motion control without any difficulties. Besides, load equalization can be turned on or off with a simple setting. With all the data

gathered from the simulations, several conclusions can be drawn, among which the following stand out.

- 1) The unified motion control keeps errors bounded and is robust to disturbances such as wind and obstacle avoidance.
- 2) The magnitude of the force acting on each of the n vehicles (f_i) in a multilift slung-load system can become significantly greater than (f_T/n) , this despite the good performance of the load equalization system with $\rho_i = (1/n)$. The global mean force decreases nonlinearly as n grows while carrying the same payload. However, if the payload weight increases in proportion to the number of vehicles, then this force remains practically constant.
- 3) In the path-following case, where the payload remains constant, the mean following error and formation errors decrease nonlinearly as n increases, and this decrease rapidly decelerates. Thus, e.g., when three URs instead of two are employed, the path-following error decreases by almost 20%, while when going from four to five, this error only decreases by 9%. This implies the need to find a relationship of commitment between the error to be tolerated and the number of URs to consider. On the contrary, when the number of URs increases in proportion to the payload weight, as in the trajectory-tracking case, the mean tracking error and formation errors remain almost constant.
- 4) There is a reduction of load oscillations as n grows. For example, when the number of URs goes from two to three, a significant decrease in load oscillations is observed in the path-following case, but there is no meaningful improvement if more than three URs are used. In the trajectory-tracking case, there is a noticeable decrease of oscillations when going from one to two vehicles, but due to the shape of the formation and the constant direction of movement, adding more vehicles does not further improve the load stabilization.
- 5) Analyzing the power consumed by the URs (Fig. 10), it can be seen that this model of mini-helicopter needs 1410 W to transport a 3-kg load and the same load with four helicopters, and each one needs 1204 W (a total of 4816 W). This is because each UR must lift its own weight, which is significantly greater than its payload capacity, and therefore, the total energy required grows in proportion to the number of helicopters used. All these considerations must be considered when searching for a balance between the energetic cost involved in the incorporation of URs into the formation, and the stability and robustness required by the mission.
- 6) The significant contribution of the proposed load equalization to multilift systems can be seen in Fig. 15. When this issue is not addressed, the performance of all tasks deteriorates (greater errors, greater oscillations, and so on), and inadmissible and dangerous situations arise, e.g., exceeding the load capacity of each UR at certain times, creating the need to add more URs to the formation to transport the same payload.

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